Meta-analysis with a single study

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Joint work with Erik van Zwet, Andrew Gelman. Available online.

The Story (funder's perspective)

- I want to fund new effective interventions but I don't want to overcommit funds
- Many interventions that people bring to me are because of selection (winner's curse)
- I can penalise the RCTs with weak evidence
- Bayesian prior is a sensible way to do it
  - Weaker evidence gets penalised more!
  - This has been proposed many times; seems to work well
- But I don't actually care about what happened in RCT, I want to know about future implementation
  - Need to consider external validity / heterogeneity
  - Worth doing a meta-analysis: estimates average effects + heterogeneity

Statistician's perspective

- Standard model for a body of research:
  - Study effect: $\beta_j = \mu + u_j$ where $u_j \sim N(0, \tau^2)$
  - Estimate: $b_j = \beta_j + \epsilon_j$ where $\epsilon_j \sim N(0, s_j^2)$
- Consider one study, $j = 1$:
  - Can't estimate between-study variation τ
  - Common practice: Ignore it! ($\tau = 0$), treating $\mu = \beta_j$
  - Results in underestimated uncertainty
  - Alternative: ignore the study (no information until at least two studies!)

Non-monotonic information gain as a function of N studies!

The Cochrane Database

- Rich source of information about treatment effects in medical trials
- 20,000+ studies publicly available
  - Mostly good quality medical trials
- Our analysis includes:
  - 1,635 meta-analyses (of 5+ studies each)
  - 18,368 trials total
  - Binary and continuous outcomes
  - Standardized to common scale (probit)
  - Let's not worry about publication bias for now
- "But what does this have to do with econ?"
  - Some databases available for econ (Brodeur et al), political science (Arel-Bundock et al), much more.
  - We analyse them in another paper.
  - You may be surprised how similar the estimates can look across databases!

Plan

- Set of sets of studies $\rightarrow$ we can estimate both μ and τ
  - there's previous literature on this
  - μ approximated well by Student's T
  - τ is log-normal
  - (μ, τ) modelled jointly would be nicer, but optional
- We can create ground truth dataset
- Use `baggr` to fit meta-analyses with single study
- Measure performance in estimating
  - β_{ij} = effect in particular study
  - μ = true population effect

What We Found in Cochrane

- Distribution of effect sizes:
  - 75% are "small" by Cohen's standards
  - Median absolute effect = 0.28
- Between-study variation:
  - Median $\tau = 0.16$
  - Roughly half the effect size

	t distribution of μ			normal of log τ	
	center	scale	df	mean	std dev
unrestricted	−0.11	0.37	5.30	−1.82	0.89
zero mean	0.00	0.37	5.18	−1.82	0.90

Table 1: *Estimated parameters of the t distribution of μ and the normal distribution of* $\log \tau$. *In the bottom row, we restrict the mean of μ to be zero.*

Simple Example

Consider a trial with:

- Measured (unbiased) effect $b = 0.7$
- Standard error $s = 0.7$
- Z-value $= 1$ (p $= 0.32$) $\rightarrow$ 95% CI: (-0.7, 2.1)

These odds look weirdly optimistic! A whole follow-up paper on that.

- Good to have a prior that shrinks toward zero to avoid overestimating effects
- There's good literature on that. (But now we want to also think about external validity!)

Zwet, Erik van, Simon Schwab, and Stephen Senn. 'The Statistical Properties of RCTs and a Proposal for Shrinkage'. *Statistics in Medicine* 40, no. 27 (2021): 6107–17. https://doi.org/10.1002/sim.9173.

Simple Example, Shrunk

```
bg <- baggr(data.frame(tau = 0.7, se = 0.7),
            prior_hypermean = student_t(5.18, 0, 0.37),
            prior_hypersd = lognormal(-1.82, 0.90))
```

After incorporating priors: - Study effect: $\hat{\beta} = 0.24$ (-0.5, 1.1) - Population effect: $\hat{\mu} = 0.17$ (-0.5, 1.0) - Shrinks toward zero! - $\hat{\beta} > \hat{\mu}$ because $\beta = \mu + u$ (extra heterogeneity means a bit more "room for shrinkage")

Inference on β vs on μ

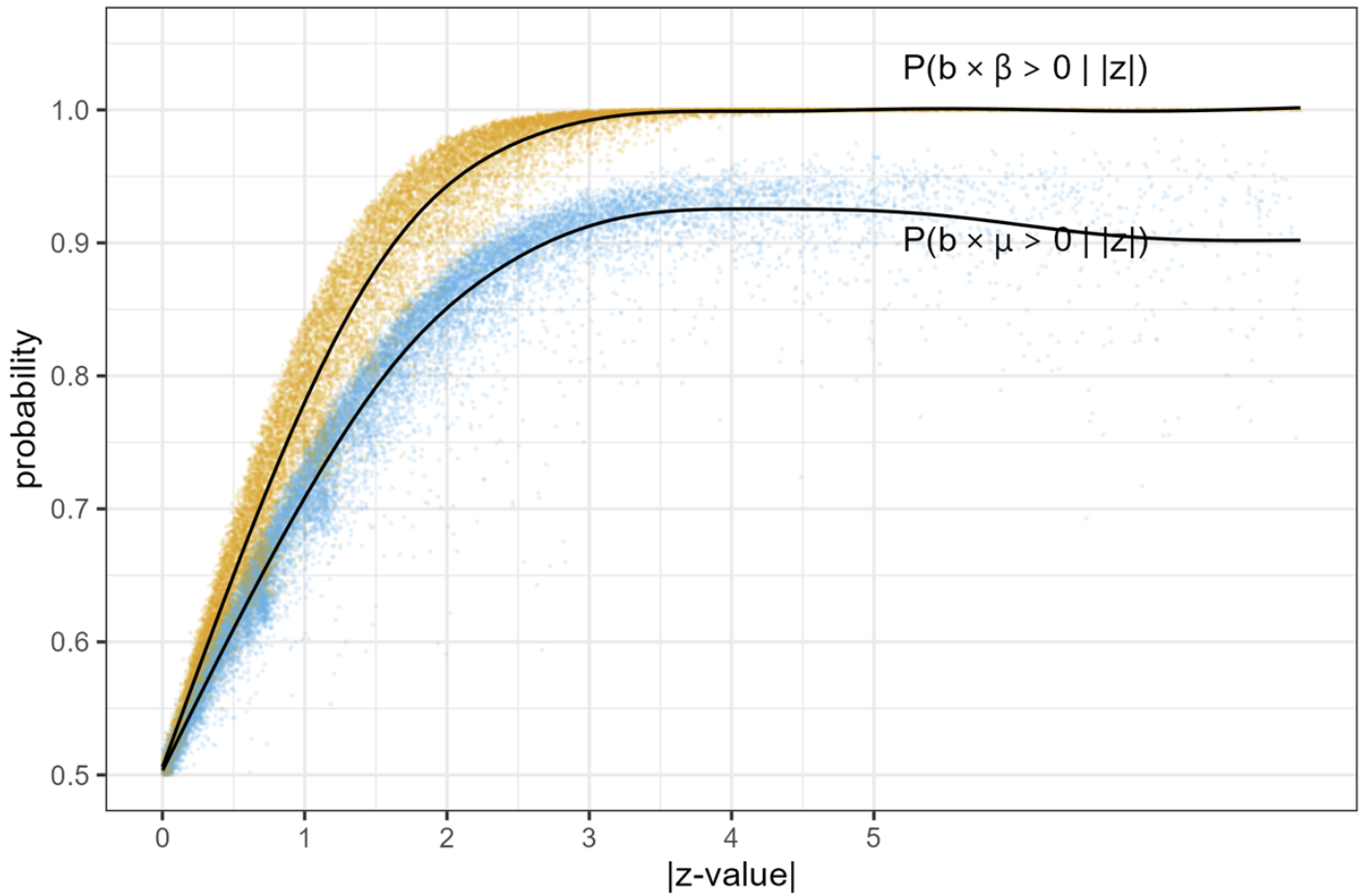


Figure 1: *Conditional probabilities of b having the same sign as β or μ given the observed absolute z-value.*

Aside: It Is In Fact A Modest Proposal

- I am not really messing with how you work
- RCT designer still gets to report their unbiased (?) estimates
- Decision maker then gives everyone the same penalty
- Fair, but not pointless: weak claims affected more
- So I am not even arguing for different or bigger trials

Synthetic data

- Now let's use Cochrane to create some synthetic trials, so that we know ground truth.
  - Realistic effect sizes, realistic errors, realistic relationship between heterogeneity and effects
- Then measure performance of β and μ estimation
  - Remember, these are two different parameters
- All of this done with `baggr`: fast and easy to use (few lines of code)

Performance Results (population-level effects μ)

- Root Mean Squared Error reduced by 34%
  - **Equivalent to doubling sample size!**
- Better coverage of uncertainty intervals
  - Traditional: 82% coverage
  - Our method: 93% coverage
- Especially valuable for significant results; reduces "winner's curse" (0.41 bias!)

	unconditional			statistically significant		
method	RMSE	bias	coverage	RMSE	bias	coverage
unbiased	0.53	0.15	0.82	0.72	0.41	0.63
Bayes	0.35	-0.11	0.93	0.38	-0.03	0.94

Table 3: *Performance of the unbiased and Bayesian estimators of the population average effect μ. The right side of the table shows the average over the statistically significant trials, i.e. $|z^*_{ij}| > 1.96$.*

Limitations and extensions

- For this paper, we do not partition heterogeneity into different dimensions
  - e.g. population, treatment, implementation
  - A real decision maker would want to
- This argument extends to K = 2, 3, 4, 5 trials.
  - The results exist for it, potential follow-up paper.
  - You can ignore priors as $K \to \infty$, but that K can be high for some problems
- Risk aversion and preference for low variance outcomes (follow-up paper)
- ???

All together now

- In social science treatment effects vary across experiments
- Experimental researchers worry about external validity
- In large sets of experiments, meta-analysis can offer some reassurance
- What if only a single trial exists?
- We can still learn from thousands of past meta-analyses *on other treatments*
- Prior information can mean effectively doubling sample size relative to naive approach (?!)
  - This matters less and less as number of trials increases
- Highly relevant to decision makers looking to fund new interventions
- You can and should try it at home